

CS 342: Computer Vision and Pattern Recognition

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Shanti Mantra

Lyrics in Sanskrit

Meaning in English

ॐ सह नाववतु।

Om, May we all be protected

सह नौ भुनक्तु।

May we all be nourished

सह वीर्यं करवावहै।

May we work together with great energy

तेजस्यि नावधीतमस्तु

May our intellect be sharpened (may our study be effective)

मा विद्विषावहै।

Let there be no Animosity amongst us

ॐ शान्तिः शान्तिः शान्तिः ॥

Om, peace (in me), peace (in nature), peace (in divine forces)

Last time

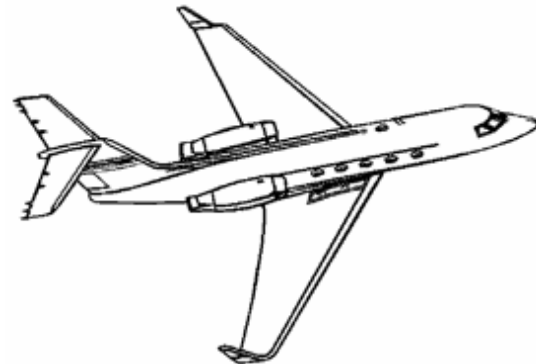
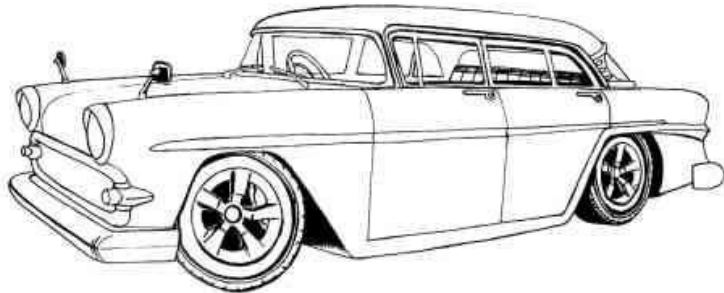
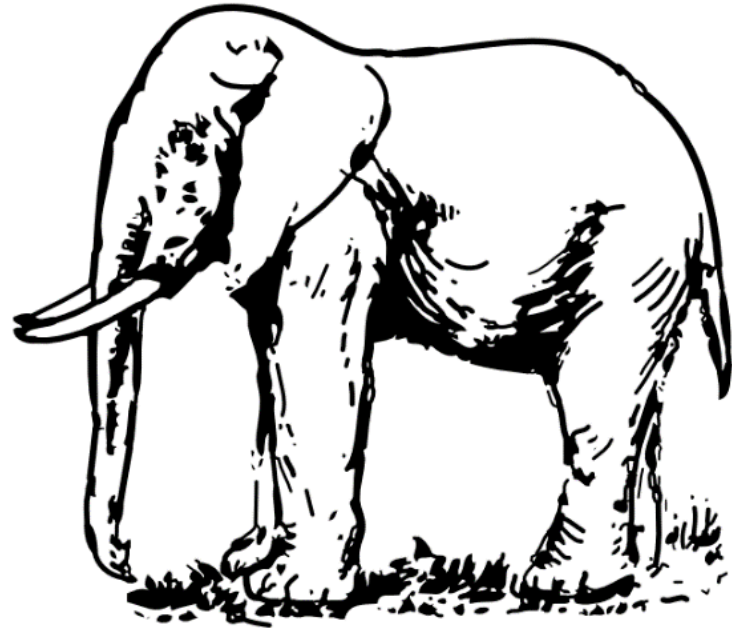
- Various models for image “noise”
- Linear filters and convolution useful for
 - Image smoothing, removing noise
 - Box filter
 - Gaussian filter
 - Impact of scale / width of smoothing filter
- Separable filters more efficient
- Median filter: a non-linear filter, edge-preserving

Image filtering

- Compute a function of the local neighborhood at each pixel in the image
 - Function specified by a “filter” or mask saying how to combine values from neighbors.
- Uses of filtering:
 - Enhance an image (denoise, resize, etc)
 - Extract information (texture, edges, etc)
 - Detect patterns (template matching)

Today

Reduced Image



Edge detection

- **Goal:** map image from 2d array of pixels to a set of curves or line segments or contours.
- **Why?**

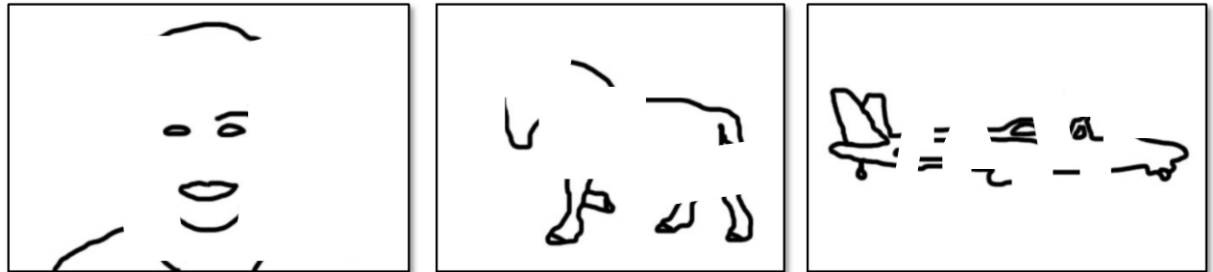
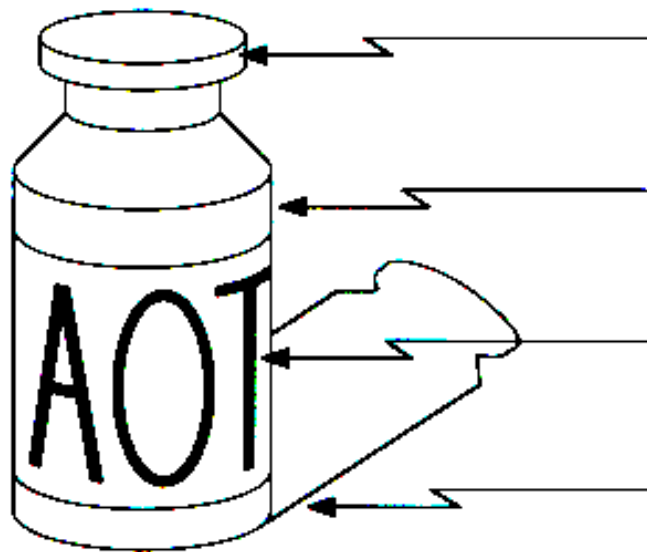


Figure from J. Shotton et al., PAMI 2007

- **Main idea:** look for strong gradients, post-process

Origin of Edges



Origin of Edges

depth discontinuity

surface color discontinuity

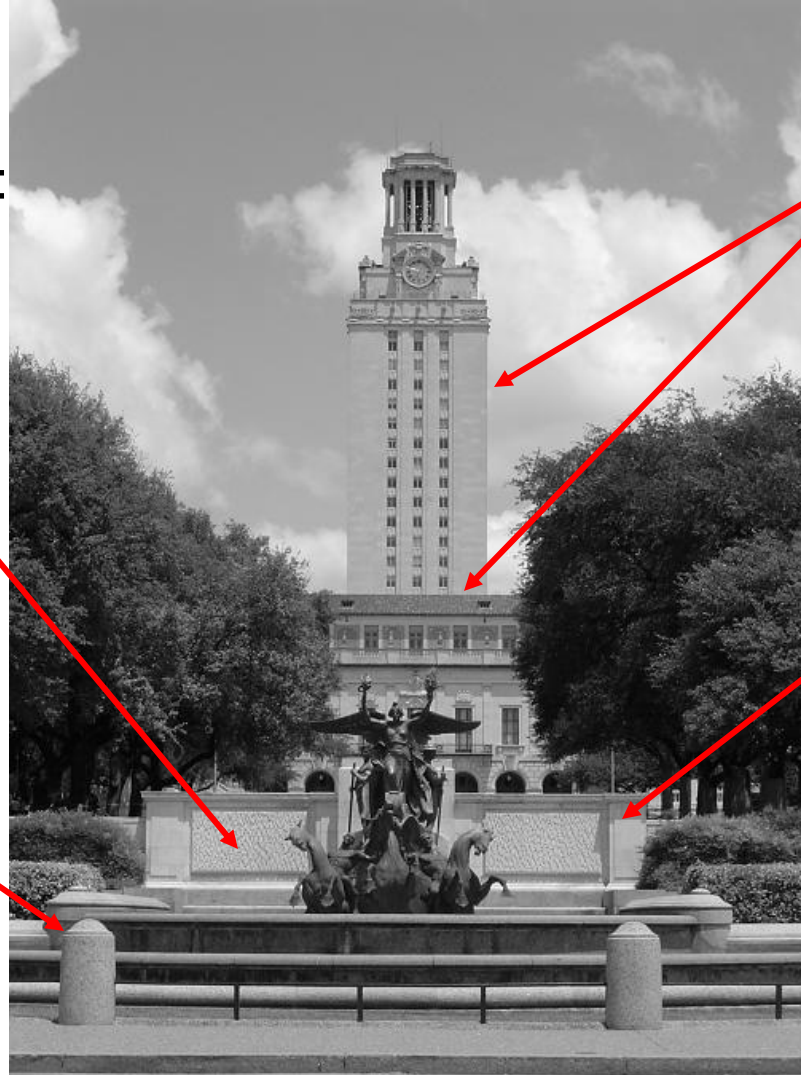
illumination discontinuity

surface normal discontinuity

What causes an edge?

Reflectance change:
appearance
information, texture

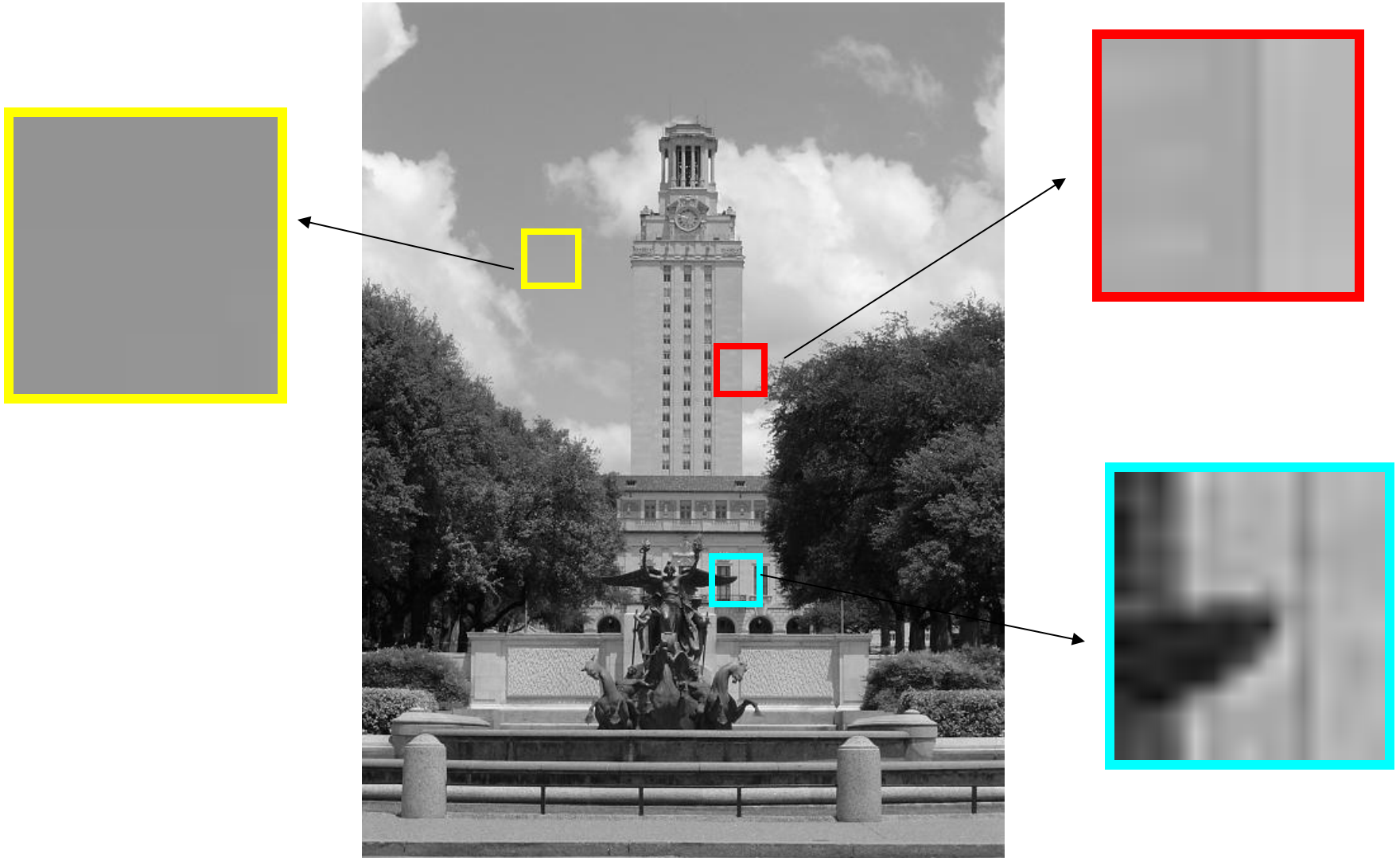
Change in surface
orientation: shape



Depth discontinuity:
object boundary

Cast shadows

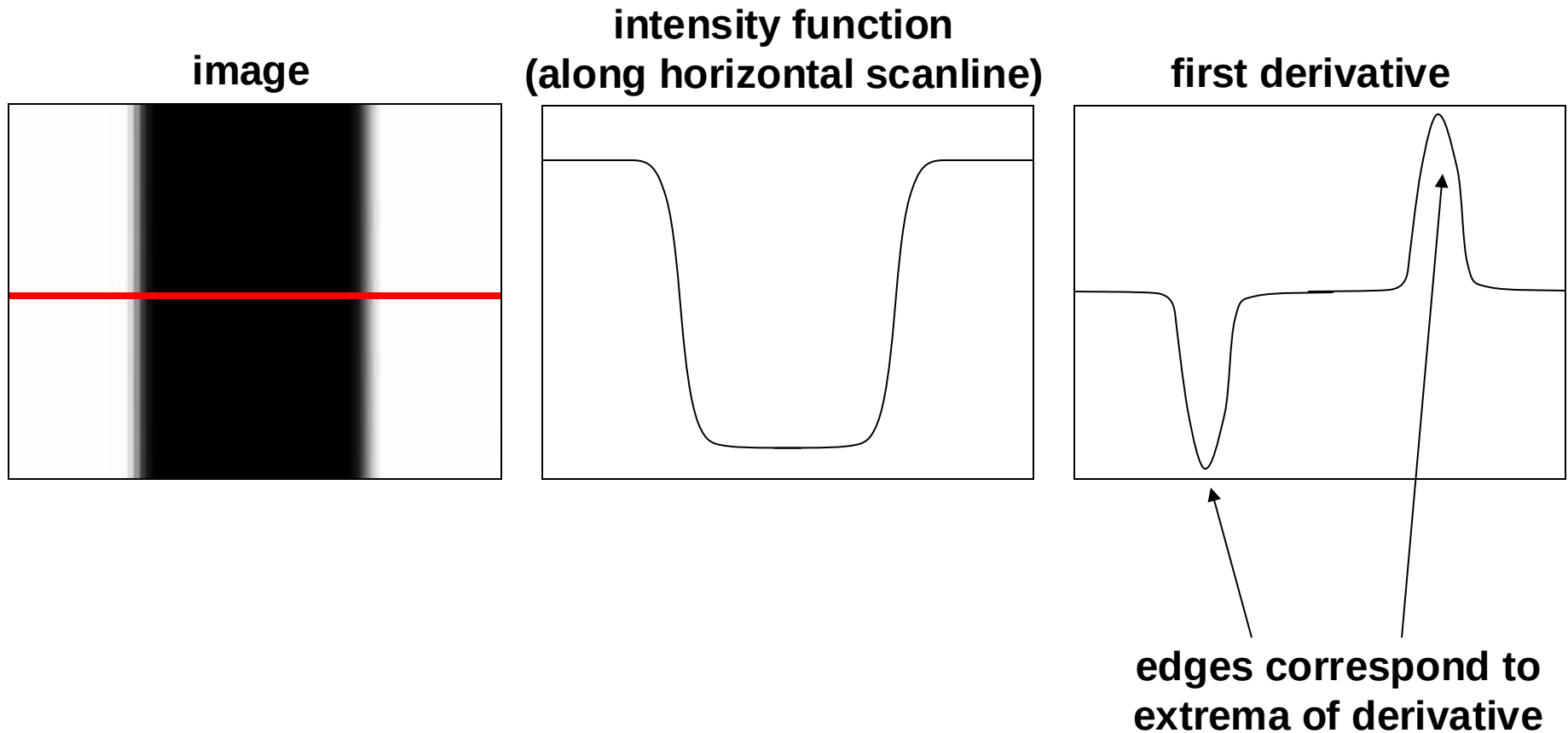
Edges/gradients and invariance



Slide credit:
Kristen Grauman

Derivatives and edges

An edge is a place of rapid change in the image intensity function.



Derivatives with convolution

For 2D function, $f(x,y)$, the partial derivative is:

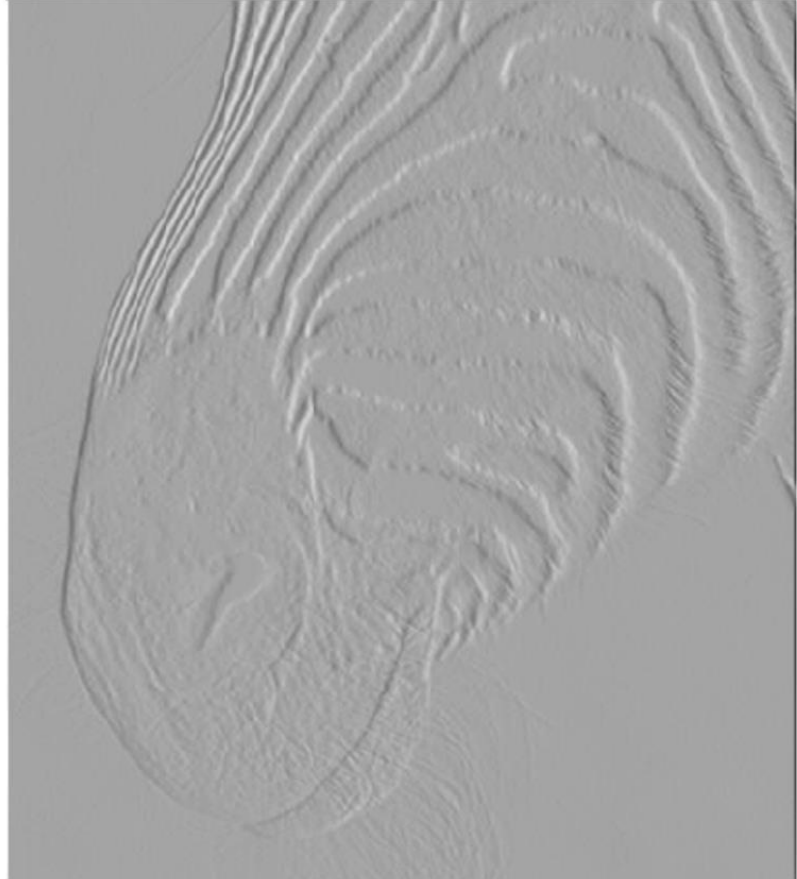
$$\frac{\partial f(x, y)}{\partial x} = \lim_{\varepsilon \rightarrow 0} \frac{f(x + \varepsilon, y) - f(x, y)}{\varepsilon}$$

For discrete data, we can approximate using finite differences:

$$\frac{\partial f(x, y)}{\partial x} \approx \frac{f(x + 1, y) - f(x, y)}{1}$$

To implement above as convolution, what would be the associated filter?

Finite differences – x or y?



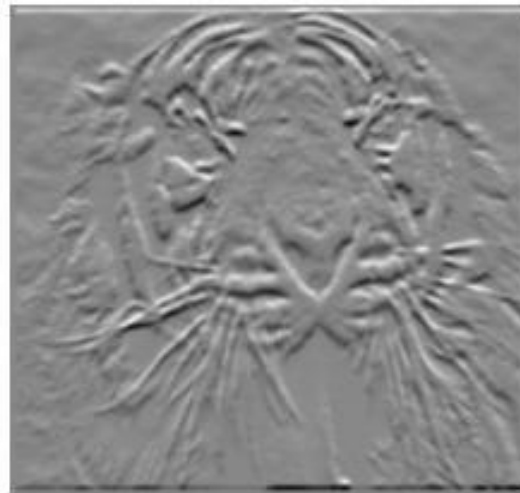
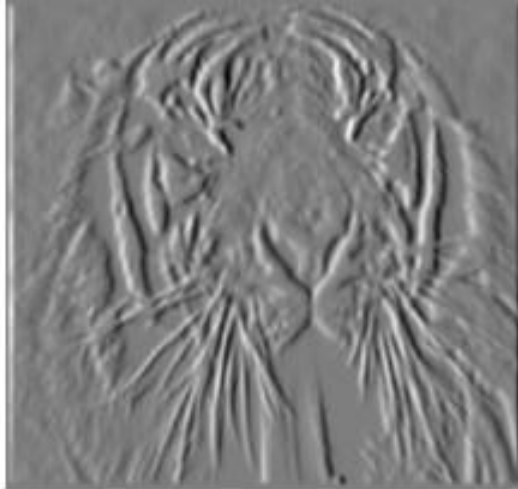
Partial derivatives of an image



$$\frac{\partial f(x, y)}{\partial x}$$

$$\frac{\partial f(x, y)}{\partial y}$$

-1	1
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-1	?	1
1	or	-1

Which shows changes with respect to x?

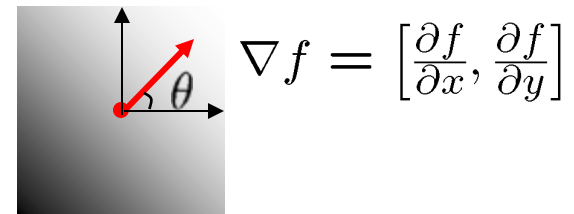
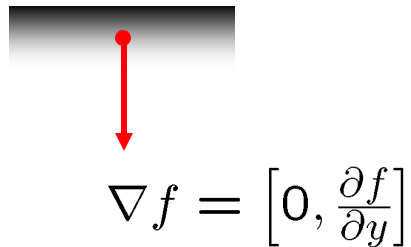
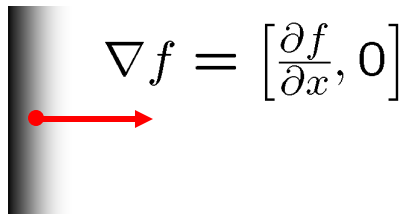
(showing filters for correlation)

Image gradient

The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

The gradient points in the direction of most rapid change in intensity



The **gradient direction** (orientation of edge normal) is given by:

$$\theta = \tan^{-1} \left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x} \right)$$

Quiz

What does it mean when the magnitude of the image gradient is zero?

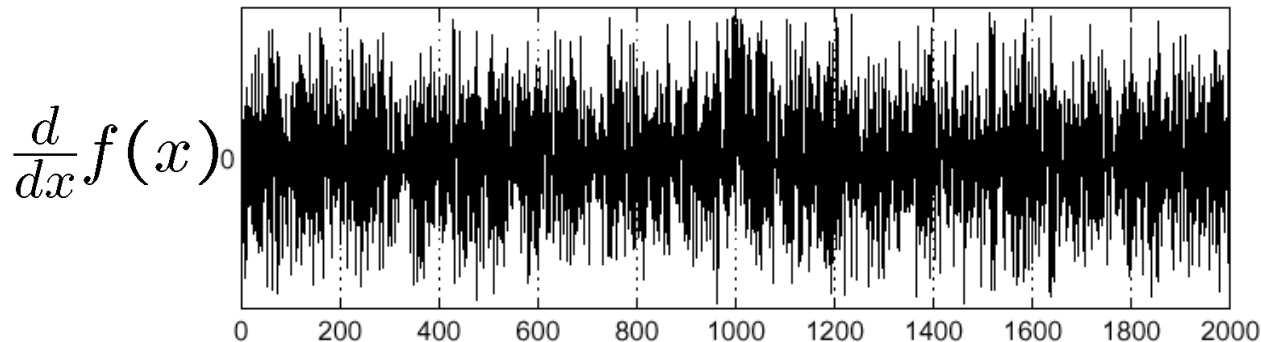
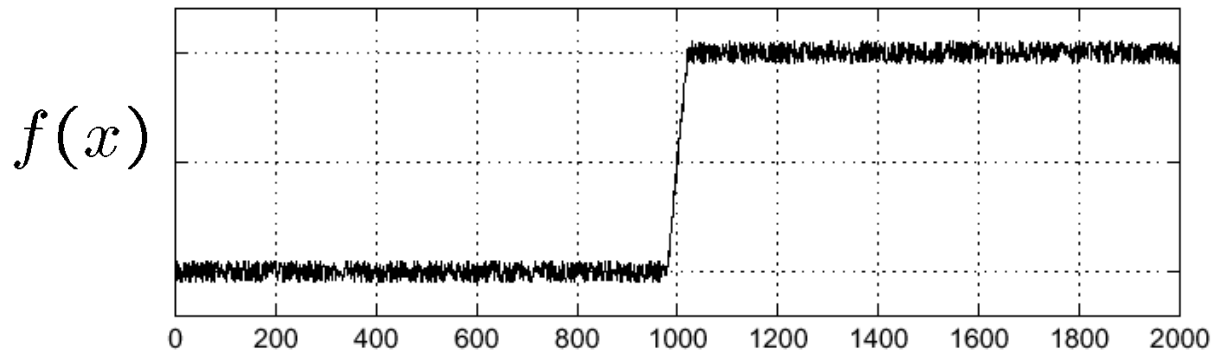
- a) The image is constant over the entire neighborhood.
- b) The underlying function $f(x,y)$ is at a maximum.
- c) The underlying function $f(x,y)$ is at a minimum.
- d) Either (a), (b), or (c).

d) Either (a), (b), or (c).

Effects of noise

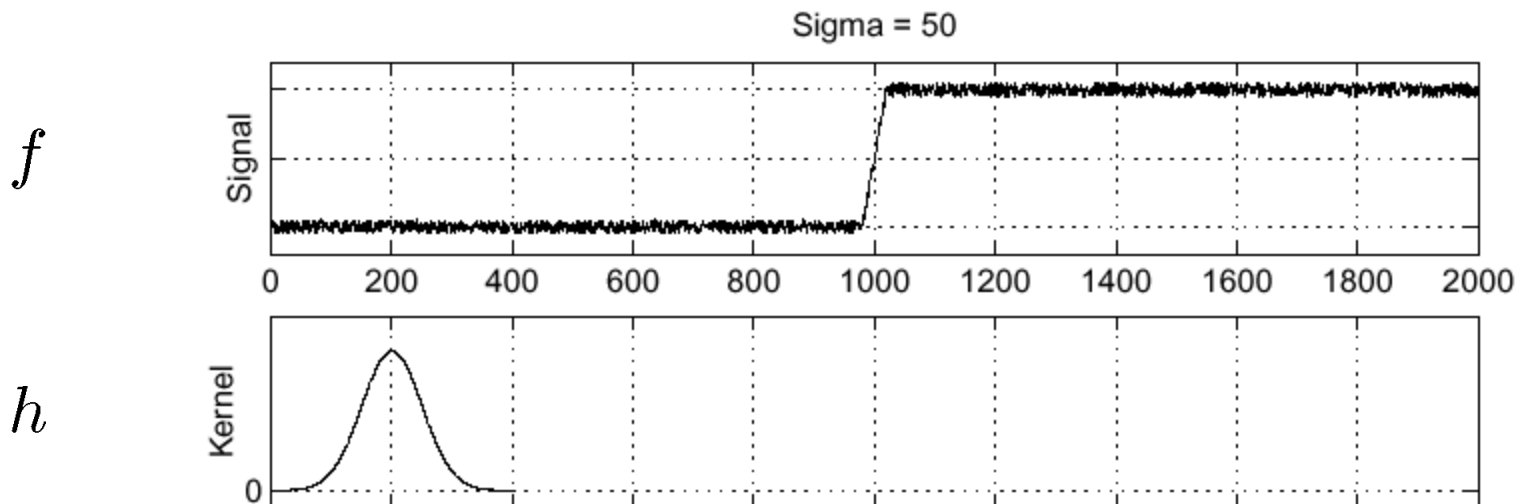
Consider a single row or column of the image

- Plotting intensity as a function of position gives a signal



Where is the edge?

Solution: smooth first



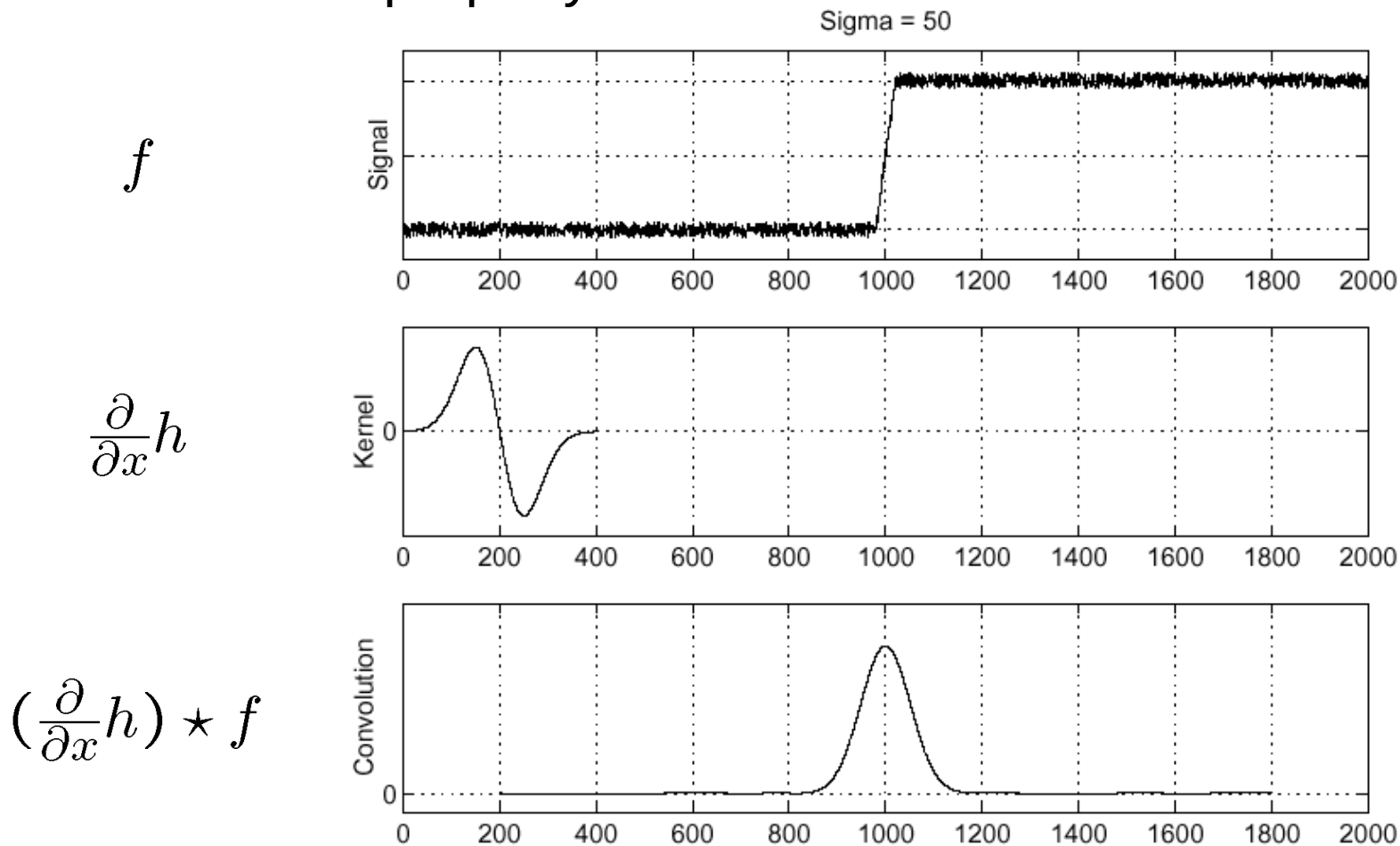
Where is the edge?

Look for peaks in $\frac{\partial}{\partial x}(h \star f)$

Derivative theorem of convolution

$$\frac{\partial}{\partial x}(h \star f) = \left(\frac{\partial}{\partial x}h\right) \star f$$

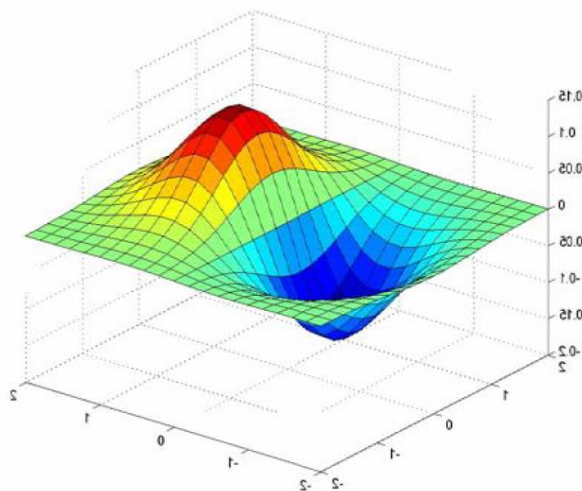
Differentiation property of convolution.



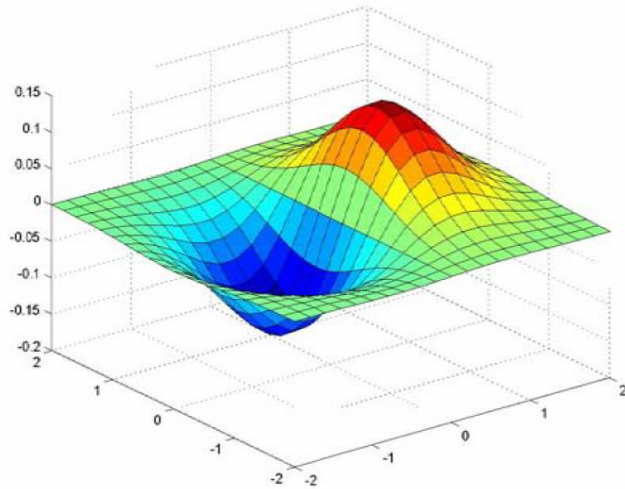
Derivative of Gaussian filters

$$(I \otimes g) \otimes h = I \otimes (g \otimes h)$$

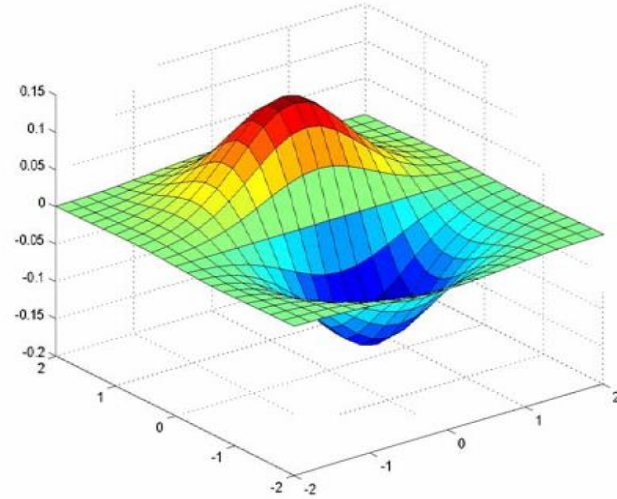
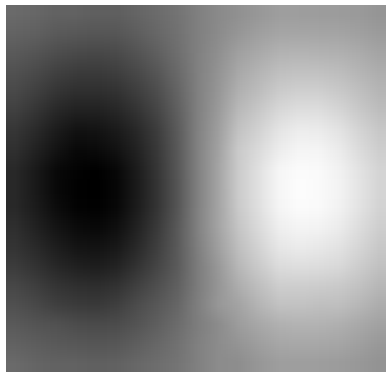
$$\begin{bmatrix} 0.0030 & 0.0133 & 0.0219 & 0.0133 & 0.0030 \\ 0.0133 & 0.0596 & 0.0983 & 0.0596 & 0.0133 \\ 0.0219 & 0.0983 & 0.1621 & 0.0983 & 0.0219 \\ 0.0133 & 0.0596 & 0.0983 & 0.0596 & 0.0133 \\ 0.0030 & 0.0133 & 0.0219 & 0.0133 & 0.0030 \end{bmatrix} \otimes \begin{bmatrix} 1 & -1 \end{bmatrix}$$



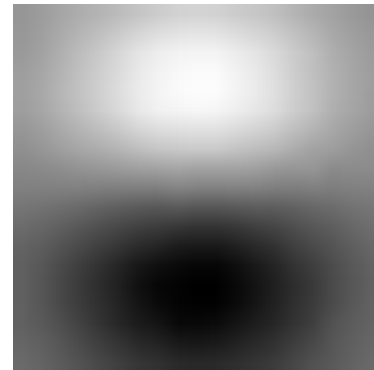
Derivative of Gaussian filters



x-direction



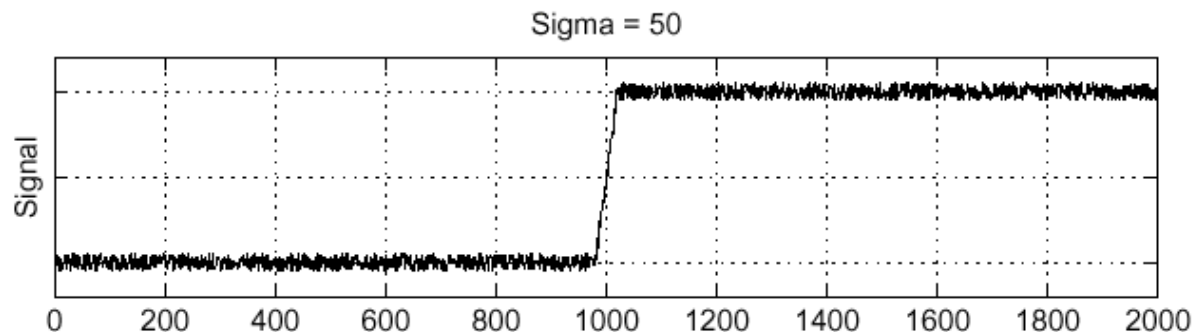
y-direction



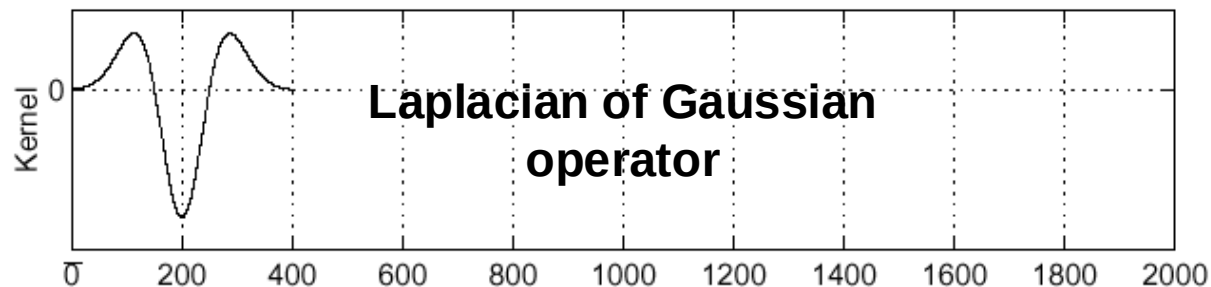
Laplacian of Gaussian

Consider $\frac{\partial^2}{\partial x^2}(h \star f)$

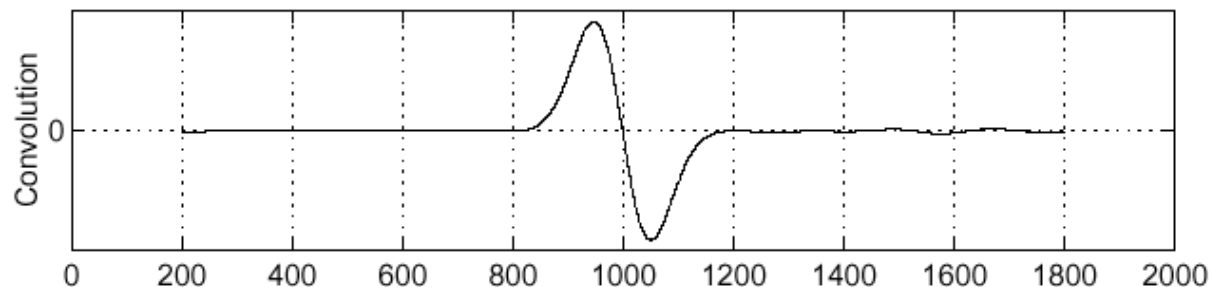
f



$\frac{\partial^2}{\partial x^2}h$



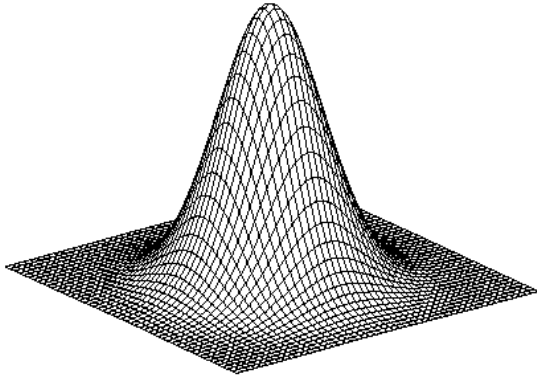
$(\frac{\partial^2}{\partial x^2}h) \star f$



Where is the edge?

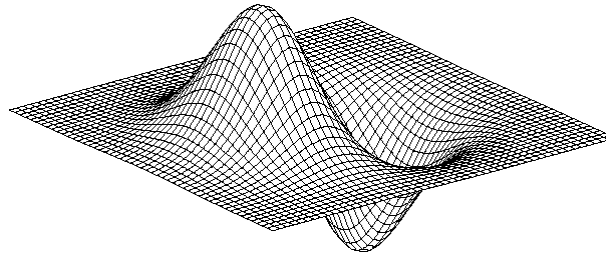
Zero-crossings of bottom graph

2D edge detection filters



Gaussian

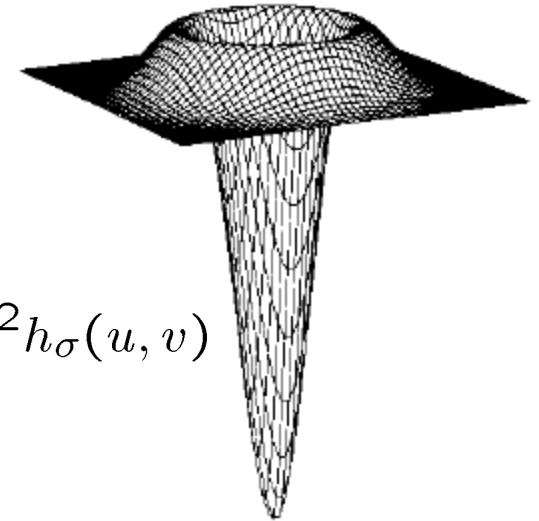
$$h_{\sigma}(u, v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$$



derivative of Gaussian

$$\frac{\partial}{\partial x} h_{\sigma}(u, v)$$

Laplacian of Gaussian



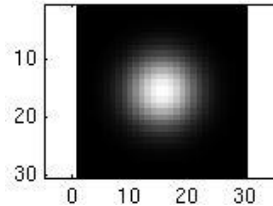
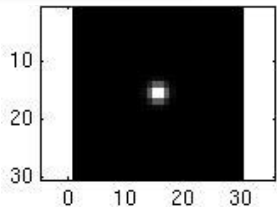
$$\nabla^2 h_{\sigma}(u, v)$$

- ∇^2 is the Laplacian operator:

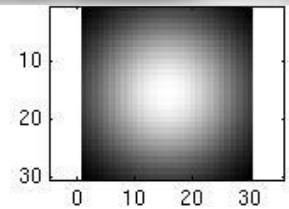
$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Smoothing with a Gaussian

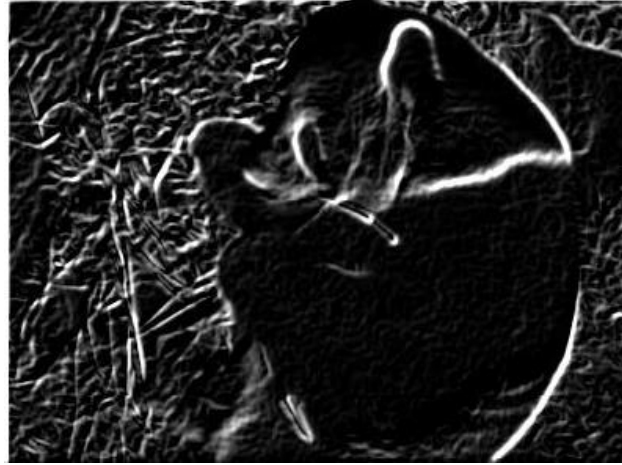
Recall: parameter σ is the “scale” / “width” / “spread” of the Gaussian kernel, and controls the amount of smoothing.



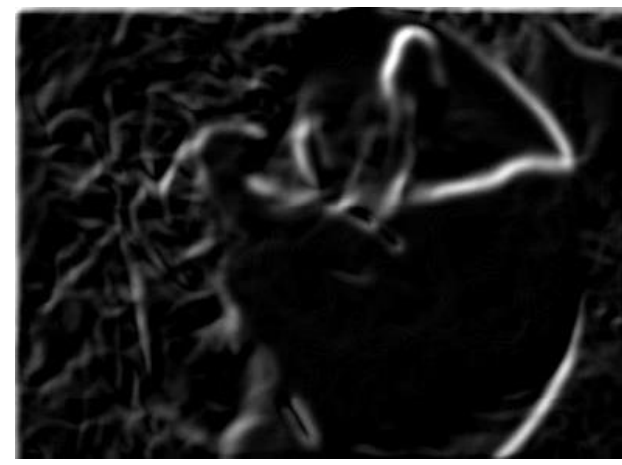
...



Effect of σ on derivatives



$\sigma = 1$ pixel



$\sigma = 3$ pixels

The apparent structures differ depending on Gaussian's scale parameter.

Larger values: larger scale edges detected
Smaller values: finer features detected

So, what scale to choose?

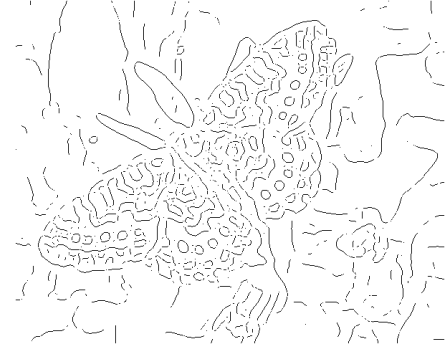
It depends what we're looking for.



Slide credit:
Kristen Grauman



Gradients $\rightarrow$ edges



Primary edge detection steps:

1. Smoothing: suppress noise
2. Edge enhancement: filter for contrast
3. Edge localization

Determine which local maxima from filter output are actually edges vs. noise

- Threshold, Thin

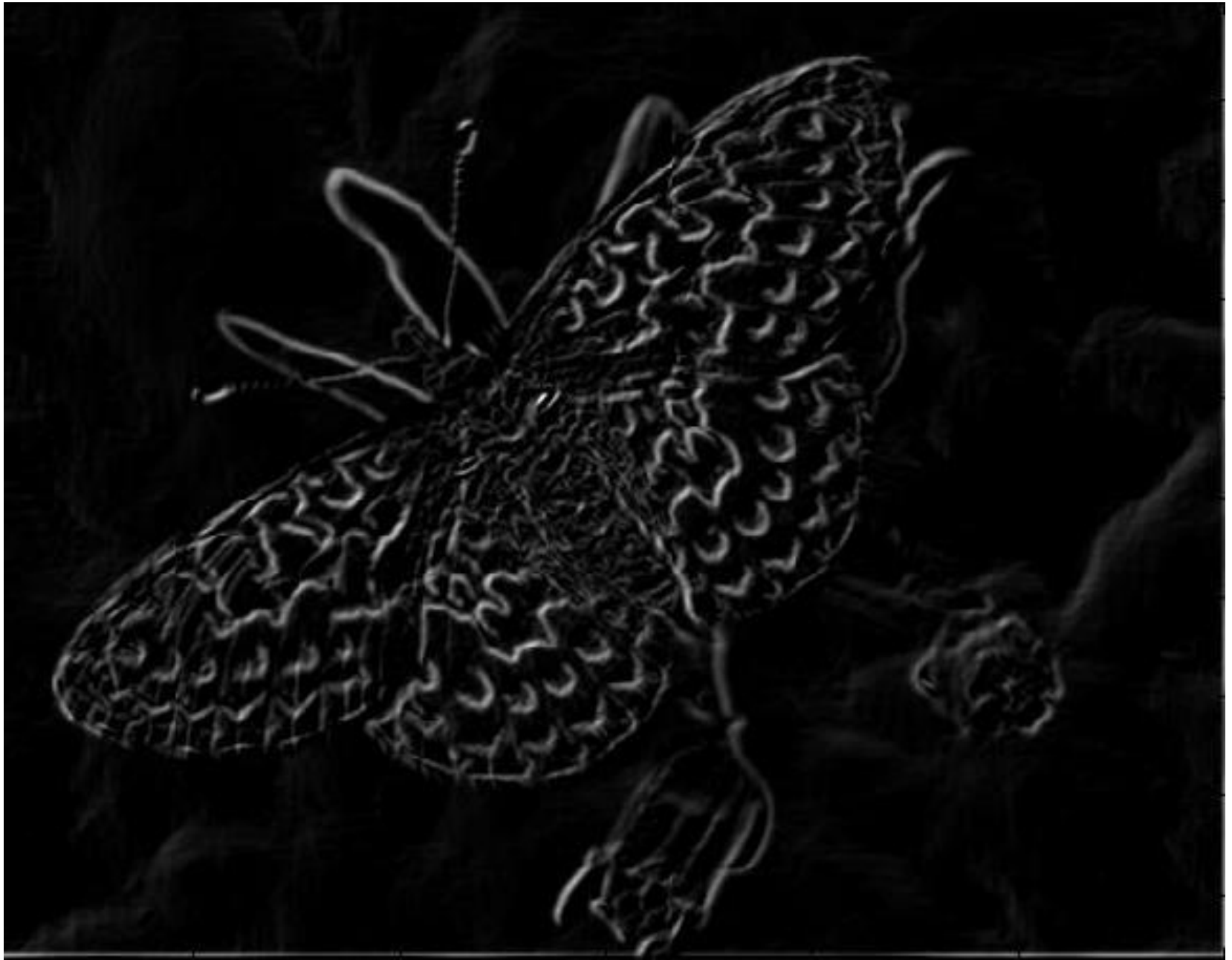
Thresholding

- Choose a threshold value t
- Set any pixels less than t to zero (off)
- Set any pixels greater than or equal to t to one (on)

Original image



Gradient magnitude image



Thresholding gradient with a lower threshold



Thresholding gradient with a higher threshold



Canny edge detector

- Filter image with derivative of Gaussian
- Find magnitude and orientation of gradient
- **Non-maximum suppression:**
 - Thin wide “ridges” down to single pixel width
- **Linking and thresholding (hysteresis):**
 - Define two thresholds: low and high
 - Use the high threshold to start edge curves and the low threshold to continue them

The Canny edge detector



original image (Lena)

The Canny edge detector



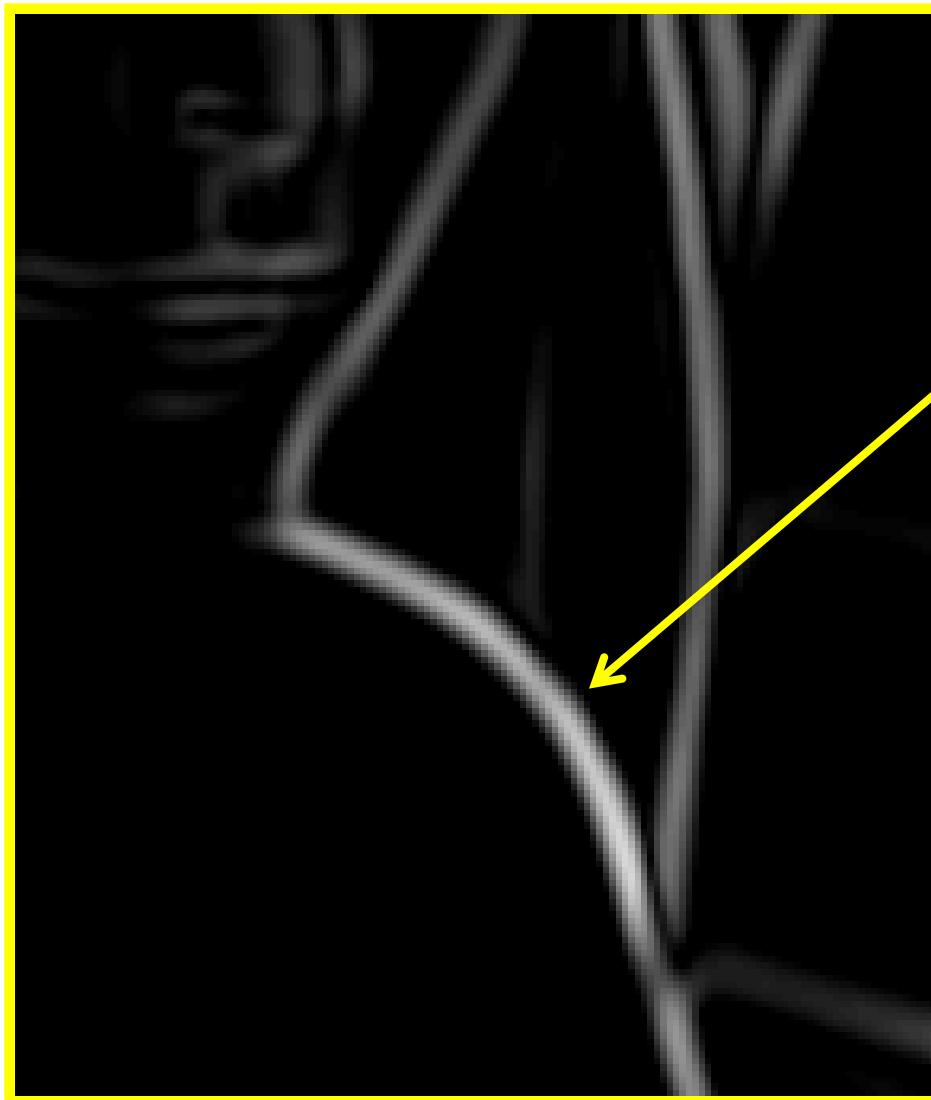
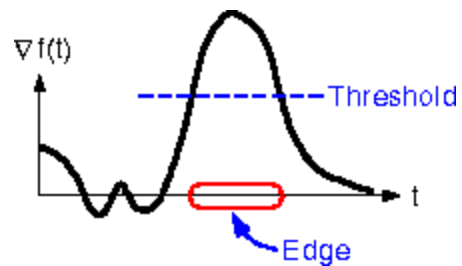
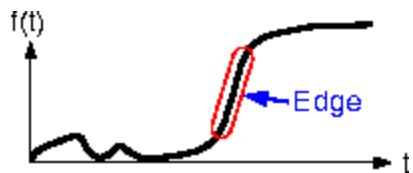
norm of the gradient

The Canny edge detector



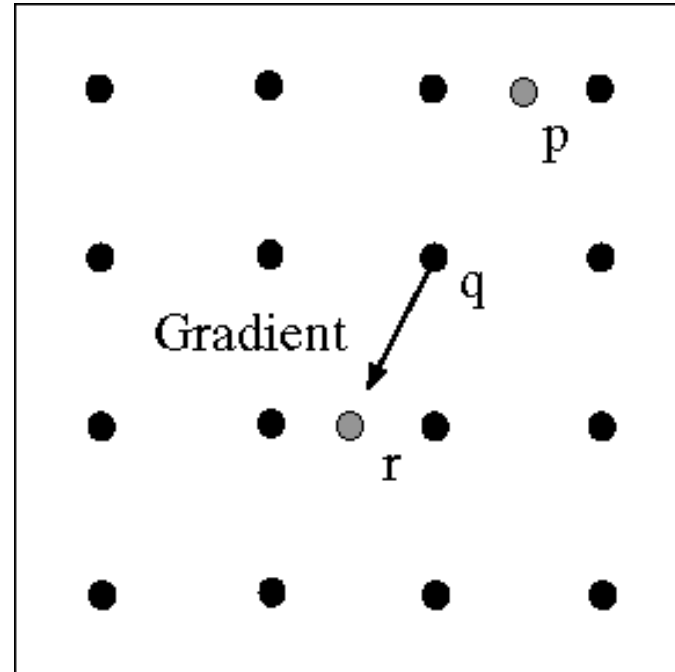
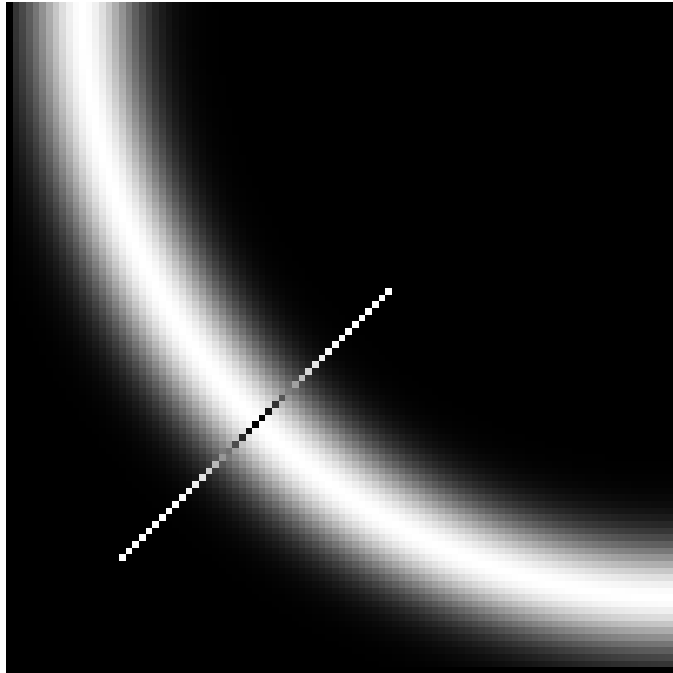
thresholding

The Canny edge detector



How to turn these thick regions of the gradient into curves?

Non-maximum suppression



Check if pixel is local maximum along gradient direction,
select single max across width of the edge

- requires checking interpolated pixels p and r

The Canny edge detector



Problem:
pixels along
this edge
didn't
survive the
thresholding

thinning
(non-maximum suppression)

Hysteresis thresholding

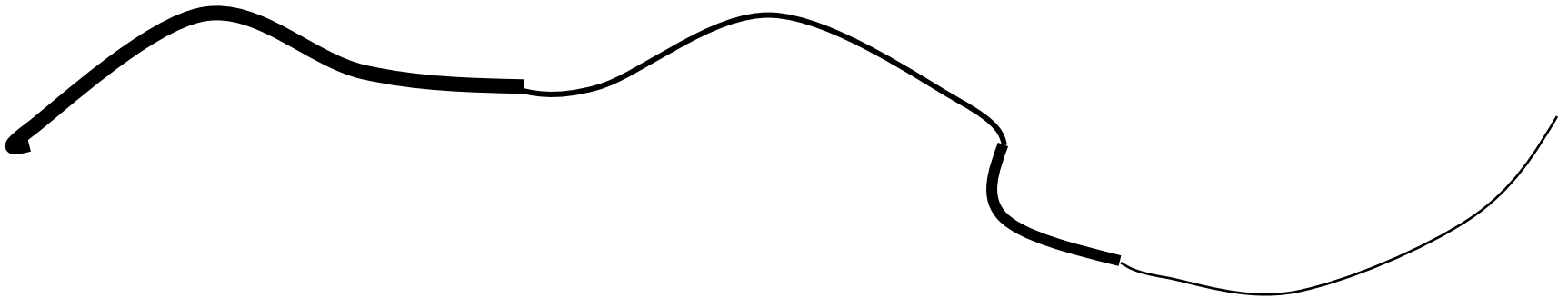
- Threshold at low/high levels to get weak/strong edge pixels
- Do connected components, starting from strong edge pixels



Credit: James Hays

Hysteresis thresholding

- Use a high threshold to start edge curves, and a low threshold to continue them.



Final Canny Edges



Credit: James Hays

Recap: Canny edge detector

- Filter image with derivative of Gaussian
- Find magnitude and orientation of gradient
- **Non-maximum suppression:**
 - Thin wide “ridges” down to single pixel width
- **Linking and thresholding (hysteresis):**
 - Define two thresholds: low and high
 - Use the high threshold to start edge curves and the low threshold to continue them

Summary

- Image gradients
- Gradients $\rightarrow$ edges and contours

Coming up

- Project 1 will be out soon.